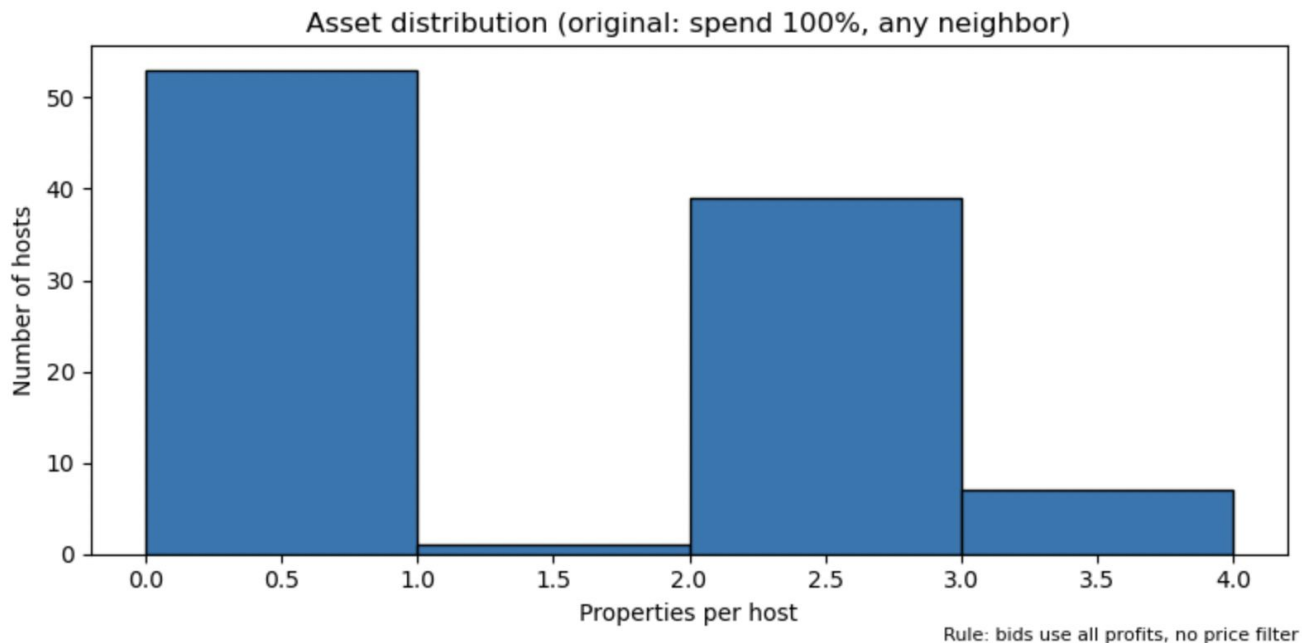


Final Project. Airbnb Market Simulation & Data Analysis

Pau Centelles, Pau Costa & Jan Rodríguez

PART 1. ADDITIONAL GRAPH

How many properties each host owns at the end of the simulation using the original rule?



Source: graph2_v0.png

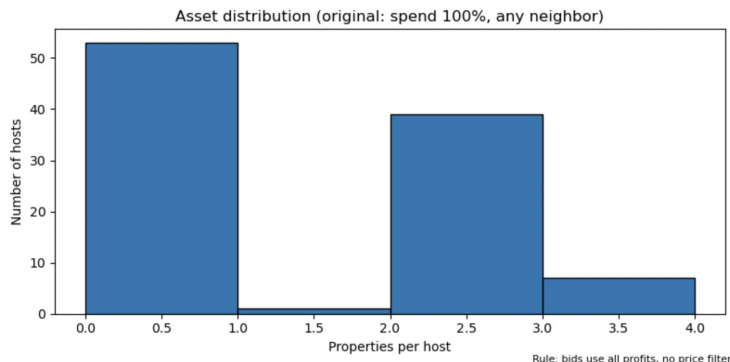
RULE CHANGE SNIPPET

Original: Bids on any neighbor if cash $\geq$ ask_price (uses 100% of profits).

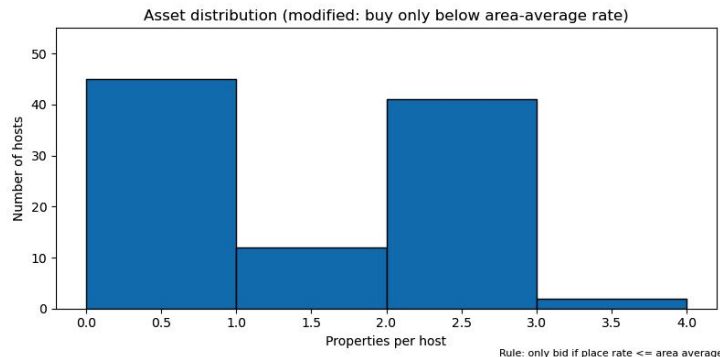
```
# Host.make_bids - original logic
if self.profits >= ask_price:
    bids.append({
        "place_id": pid,
        "seller_id": place.host_id,
        "buyer_id": self.host_id,
        "spread": self.profits - ask_price,
        "bid_price": self.profits,
    })
```

Modified: Only bids on neighbors $\leq$ the area's average rate; ignores overpriced neighbors.

```
# Host.make_bids - only bid on "cheap" neighbors
area_mean_rate = self.city.area_mean_rates[place.area]
if place.rate > area_mean_rate:
    pass # skip expensive neighbor
else:
    budget = self.profits * self.bid_spend_fraction # here 1.0
    if budget >= ask_price:
        bids.append({
            "place_id": pid,
            "seller_id": place.host_id,
            "buyer_id": self.host_id,
            "spread": budget - ask_price,
            "bid_price": budget,
        })
```



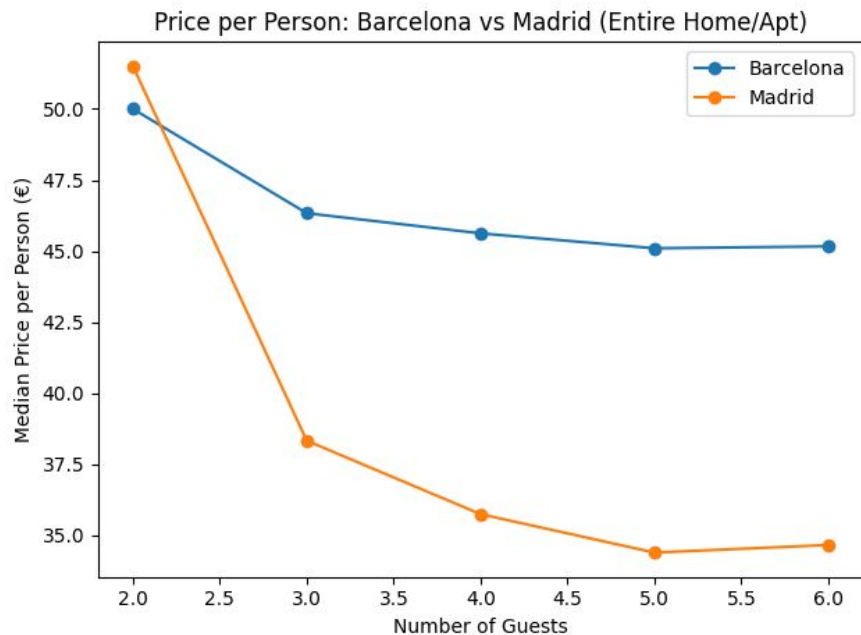
Expected: avoid overpaying $\rightarrow$ more hosts afford a purchase; lower concentration.



Observed: fewer hosts at 0, more at 1–2; 3+ tail almost disappears.

PART 2. GRAPH

How does price per person change with group size in Barcelona vs Madrid for entire homes?



Source: graph3.png

THE END

THANKS FOR ATTENDING